



Considerations for deriving a safe intake of propylene glycol

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ARTICLE INFO

Handling Editor: Dr. Bryan Delaney

ABSTRACT

The use of propylene glycol (PG) in food and other applications is widespread, and some estimates of dietary exposure to PG approach or exceed the Acceptable Daily Intake (ADI) of 25 mg/kg bw-day. The current ADI for PG applies a cumulative uncertainty factor of 100, which includes factors of 10 for both interspecies and intraspecies differences. Available toxicology studies and human data, however, indicate a plausible mode of action (MoA) that would support a chemical-specific adjustment factor (CSAF) of 1 for interspecies toxicodynamic differences, reducing the total uncertainty factor from 100 to 40. The MoA involves an increase in serum PG concentrations after metabolic saturation, leading to serum hyperosmolarity, which can ultimately cause hemolytic changes and red blood cell damage. Therefore, the species similarities in toxicodynamic response for this critical effect could support increasing the ADI from 25 to 62.5 mg/kg bw-day, applicable to both children and adults.

1. Introduction

Propylene glycol (PG), a low molecular weight diol alcohol, is a useful solvent with several industrial, consumer and food applications. PG is used in many products, such as cosmetics and personal care products, pharmaceuticals, de-icing fluids, paints, coatings, and liquid detergents (NTP, 2004; European Medicines Agency, 2017). PG is also a flavoring adjuvant in food (EFSA, 2018). PG has a critical role as a handling agent in flavoring systems (especially for foods that must meet halal certification) and is a better alternative to ethanol from a flammability and transportation standpoint. The European Food Safety Association (EFSA) estimated mean dietary PG exposures ranging from 0.1 mg/kg bw-day (infants) to 11.6 mg/kg bw-day (toddlers) across age groups among Europeans (EFSA, 2018). A high-end, 95th percentile estimate of 23.3 mg/kg bw-day for children 3–9 years represented the highest dietary intake among the age groups examined, measured using a conservative brand-loyal scenario. In the US, the only intake estimates come from a relatively crude evaluation method based on the total annual production/use of PG. In these evaluations, daily PG exposure was estimated to be 14–34 mg/kg bw-day for a 70-kg adult (NTP, 2004; US EPA, 2008) (see Discussion for more information on PG exposure estimates). JECFA also reviewed the use of PG as a flavoring adjuvant in 2001 and concluded no safety concerns at estimated daily dietary

exposures of 2400 mg per capita (Joint FAO/WHO Expert Committee on Food Additives, 2002).

Several agencies have derived safe levels for PG in foods. The Joint Food Agriculture Organization (FAO)/World Health Organization (WHO) Expert Committee on Food Additives (JECFA) defines the acceptable daily intake (ADI) as an amount of an additive in food or drinking water that can be safely consumed daily over a lifetime without adverse health effects (World Health Organization, 2023). In 1974, JECFA established an ADI of up to 25 mg/kg bw-day, which was reaffirmed by EFSA in 2018 (Joint FAO/WHO Expert Committee on Food Additives, 1974; EFSA, 2018). The ADI is derived by applying uncertainty factors (UF) to the most sensitive point of departure (POD) from relevant animal studies (IPCS, 2005; EFSA, 2018). The existing ADI and most other health-based guidance values (HBGVs) for PG are derived from rat or dog studies conducted in the 1970s (e.g., Gaunt et al., 1972; Weil et al., 1971) (Table 1). The existing ADI of 25 mg/kg bw-day is based on a no observed adverse effect level (NOAEL) and POD of 2500 mg/kg bw-day PG from dietary administration to rats for two years (EFSA, 2018; Gaunt et al., 1972). An uncertainty factor of 100 (i.e., 10 for interspecies differences, 10 for intraspecies variability) was applied. A summary of HBGVs for PG, along with other relevant information, is provided in Table 1.

In re-evaluating the existing ADI, we reviewed the scientific

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<https://doi.org/10.1016/j.fct.2024.114460>

Received 6 November 2023; Received in revised form 5 January 2024; Accepted 14 January 2024

Available online 5 February 2024

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literature on PG toxicity including animal studies, human clinical data, and toxicokinetic data, to determine whether a revision was justified. In particular, our analysis focused on intra- and interspecies differences in toxicodynamics and toxicokinetics, and whether the data supported the use of chemical-specific adjustment factors (CSAFs) for these areas of uncertainties.

2. Methods

We conducted literature searches for PG toxicology information, including toxicokinetic and toxicodynamic studies in humans, animals, and *in vitro*, as well as human clinical studies. PubMed and Scopus were utilized to identify clinical and laboratory studies published through October 2022. We focused our search on studies of oral exposure. To identify existing comprehensive assessments, ToxPlanet (an online database that compiles all safety assessment documents for chemicals from national and international regulatory and research agencies) was searched. In cross-referencing the bibliographies of review articles and assessment documents such as EMA (2017) and EFSA (2018), other potentially relevant studies were identified.

For human studies, we used the following search terms:

((("propylene glycol" OR "1,2-dihydroxypropane" OR "1,2-propanediol" OR "methyl glycol" OR "trimethyl glycol") OR (57-55-6[EC/RN Number])) AND ("oral expos*" OR "oral* administrat*" OR "administration, oral"[Mesh] OR "oral intake" OR "oral ingestion") AND (ADME OR absorption OR distribution OR metabolism OR excretion) AND (Pharmacokinetics[Mesh] OR Pharmacokinetics[Subheading] OR pharmacokinet* OR toxicokinet*) AND ("2018/01/01"[Date - Publication] : "2022/10/13"[Date - Publication]) NOT ("oral bioavailability"))

For animal and *in vitro* studies, we used the following search terms:

((("propylene glycol" OR "1,2-dihydroxypropane" OR "1,2-propanediol" OR "methyl glycol" OR "trimethyl glycol") OR (57-55-6[EC/RN Number])) AND ("oral expos*" OR "oral* administrat*" OR "administration, oral"[Mesh] OR "oral intake" OR "oral ingestion") AND (toxicity OR toxicolog* OR toxicology OR toxic* OR carcinogen* OR Carcinogenesis OR cardiotox* OR Cardiotoxicity OR Toxicokinetics OR carcinogenicity OR embryoletality OR embryotoxicity OR

Table 1

A summary of health-based guidance values for propylene glycol.

Agency (Year)	POD	UFs	HBGV (mg/kg bw-day)	Contextual Information
Joint FAO/WHO Expert Committee on Food Additives, 1974)	NOAEL = 2500 mg/kg bw-day 2-year diet study in rats (Gaunt et al., 1972)	Total = 100 10 (interspecies) 10 (intraspecies)	ADI = 25	• No significant differences in hematological parameters (e.g., RBC and white blood cell counts), urinary and renal examinations, organ weights, cumulative death rates, mean body weights and food intake for all PG exposure groups.
US EPA (2008)	LOAEL = 5200 mg/kg bw-day 5-week drinking water study in rats (Vaille et al., 1971; as cited in US EPA, 2008)	Total = 300 3 (LOAEL to NOAEL) 10 (interspecies) 10 (intraspecies)	PPRTV = 20	• Based on reduced RBC counts and hyperglycemia in rats.
EMA (2017)	NOAEL = 5000 mg/kg bw-day 2-year diet study in dogs (Weil et al., 1971)	Total = 100 2 (interspecies) 10 (intraspecies) 5 (duration)	PDE = 50 (adult)	• PDE from non-clinical data. • EMA concluded that observed hematological changes at 5000 mg/kg bw-day were not statistically significant and within normal biological ranges. • NOAEL derived according to ICH guidelines.
EMA (2017)	NOAEL = 1000 mg/kg bw-day Single IP exposure in juvenile mice (Lau et al., 2012)	Total = 1200 12 (interspecies) 10 (intraspecies) 10 (duration)	PDE = 1 (children, <5 years of age)	• PDE from non-clinical data. • Findings of apoptosis in a neurodevelopmental study not an appropriate basis for POD in view of single IP injection. • NTP's continuous breeding study (18-week) found no effects on reproduction or development (NTP, 2004). • NOAEL derived according to ICH guidelines.
EMA (2017)	None	None	Safety Limit = 500 (adult)	• Safety limit from clinical data. • Doses up to 500 mg/kg bw-day PG could be administered to adult patients for long-term periods with no adverse effects (Yaucher et al. 2003; Yahwak et al. 2008).
EMA (2017)	None	None	Safety Limit = 50 (children, <5 years of age)	• Safety limit from clinical data. • Based on human equivalent dose (HED) of the mouse NOAEL (Lau et al., 2012). • No short-term biochemical effect from median PG exposure of 34 mg/kg bw-day in 69 pre-term and low birth weight infants (Allegaert et al., 2010).
EMA (2017)	None	None	Safety Limit = 1 (children, <1 month of age)	• Safety limit based on clinical data. • Low body PG clearance in neonates (De Cock et al., 2014).
EFSA (2018)	NOAEL = 2500 mg/kg bw-day 2-year diet study in rats (Gaunt et al., 1972)	Total = 100 10 (interspecies) 10 (intraspecies)	ADI = 25	• Reaffirmed Joint FAO/WHO Expert Committee on Food Additives, 1974) ADI. • Applicable to adults and children >12 weeks.

Notes:

ADI = Acceptable Daily Intake; bw = Body Weight; EFSA = European Food Safety Authority; EMA = European Medicines Agency; US EPA = United States Environmental Protection Agency; HBGV=Health-Based Guidance Value; HED=Human Equivalent Dose; ICH = Internal Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use; ICPS = International Programme on Chemical Safety; IP = Intraperitoneal; JECFA = Joint FAO/WHO Expert Committee on Food Additives; LOAEL = Lowest Observed Adverse Effect Level; NOAEL = No Observed Adverse Effect Level; NTP=U.S. National Toxicology Program; PDE = Permitted Daily Exposure; PG = Propylene Glycol; POD=Point of Departure; PPRTV = Provisional Peer-Reviewed Toxicity Value; RBC = Red Blood Cell; UF = Uncertainty Factor.

Teratogenicity OR Terato* OR mutagenesis OR Aneuploidy OR polyploidy OR neurotox* OR nephrotox* OR immunotox* OR hepatotox* OR Epigenetic* OR genotox* OR Cytotox*) AND ("2018/01/01"[Date - Publication] : "2022/10/13"[Date - Publication])

For human clinical studies, we used the following search terms:

("propylene glycol" OR "1,2-dihydroxypropane" OR "1,2-propanediol" OR "methyl glycol" OR "trimethyl glycol") OR (57-55-6[EC/RN Number])) AND ("oral expos*" OR "oral* administrat*" OR "administration, oral"[Mesh] OR "oral intake" OR "oral ingestion") AND ("Clinical Studies"[Publication Type] OR "clinical study"[Text Word] OR "clinical trial"[text word] OR "clinical trials"[Publication Type]) AND ("2018/01/01"[Date - Publication] : "2022/10/13"[Date - Publication]) NOT (animal* OR rat OR rats OR mice OR mouse)

We reviewed the available animal toxicity studies to identify the critical effect, the key study, and a POD to form the basis of an updated ADI. We assessed the reliability of animal studies using the Klimisch scoring system (Klimisch et al., 1997). We only considered studies with K = 2 or higher for POD selection. A qualitative comparison of interspecies and intraspecies toxicokinetics and toxicodynamics was performed. In accordance with guidelines from the International Programme on Chemical Safety (IPCS), we considered whether the data supported the use of CSAFs to address inter- and intraspecies differences. Relevant data from human clinical studies were used to support refinements to the toxicodynamic CSAF.

In most of the human studies, PG dose in mg/kg bw-day was calculated using assumptions regarding body weight, lorazepam infusion rates (PG was used as an excipient in lorazepam preparations), and exposure duration. Lorazepam is a benzodiazepine that is used to induce a sedative effect in patients. As such, a neurological assessment was not often not possible in the human studies (European Medicines Agency, 2017). Unless otherwise noted, instead of presenting all of the PG dose calculation assumptions in this manuscript, we confirmed and relied on the study-specific doses presented in Appendix 3 of European Medicines Agency (2017).

3. Results

3.1. Animal studies

The toxicity of PG has been well studied in several animal models, such as rats, mice, dogs, and monkeys (European Medicines Agency, 2017; Thackaberry et al., 2010), including two-year bioassays in rats (Gaunt et al., 1972) and dogs (Weil et al., 1971), a continuous breeding study in mice (NTP, 2004), an acute study in rats (Saini et al., 1996) and a 90-day study in monkeys and other species (Thackaberry et al., 2010) (Table 2). Based on studies in animals, NTP stated there was "clear evidence" that PG does not cause developmental or reproductive toxicity (NTP, 2004). The critical effect and POD would have to be determined from available sub-chronic and chronic toxicity studies. While most studies reported no effects, some observations were noted in the two-year dog study (Weil et al., 1971).

Effects on red blood cells (RBCs) were observed at the highest dose in the 2-year study in Beagle dogs (Weil et al., 1971) considered to be the critical effect by Joint FAO/WHO Expert Committee on Food Additives, (1974) and EFSA (2018). In this study, dogs were administered PG dietary doses of 2000 and 5000 mg/kg bw-day (Weil et al., 1971). Blood chemistry, hematology, and urine analysis were measured prior to the start of treatment, and after 6, 12, and 23 months of treatment. All dogs were necropsied at the end of the study, and gross and histopathologic examinations of organs were conducted. For an isocaloric control, dextrose was administered at doses equivalent to 2540 and 6350 mg/kg bw-day. Hematological parameters showed dose-dependent, statistically significant changes, which were small in magnitude but indicative of increased hemolysis and a compensatory increase in RBC production

(Table 3). In the high-dose group (5000 mg/kg bw-day), total erythrocytes and hemoglobin were marginally reduced compared to controls. This change occurred in both male and female dogs, but effects were more pronounced in females. Females in the high-dose group also exhibited slight changes in percentage of hematocrit, percentage of nucleated erythrocytes (RBCs), and percentage of reticulocytes. These changes were mild, mostly within normal limits, and, therefore, were not considered clinically significant. However, JECFA (1974) and EFSA (2018) each considered the hematological findings to be adverse and identified a NOAEL of 2000 mg/kg bw-day and LOAEL of 5000 mg/kg bw-day. In contrast, EMA (2017) did not consider these changes to be adverse and set the NOAEL at 5000 mg/kg bw-day, which is consistent with the interpretation by the original study authors.

Hematologic changes have also been observed in rat studies. When PG was given to Wistar female rats as a single bolus dose of 700 or 2940 mg/kg bw via gavage, hematologic effects including a decrease in hemoglobin, packed cell volume, and RBC counts were observed at both doses (Saini et al., 1996). According to EMA (2017), the bolus exposure was responsible for the effects, as similar doses in feed were not associated with any effects on erythrocytes over a much longer study period (1 day vs. 2 years). In a 5-week drinking water study in rats (Vaille et al., 1971; as cited in US EPA, 2008), reduced RBC counts at 5200 mg/kg bw-day were also noted, the latter informed US EPA's Provisional Peer-reviewed Toxicity Value (PPRTV) (US EPA, 2008).

In a 2-year rat study (Gaunt et al., 1972), PG was administered at 0; 6250; 12,500; 25,000 or 50,000 ppm in feed. Blood chemistry, hematology such as changes to RBC and white blood cell (WBC) counts, and urine analysis were measured throughout the study. No significant differences were noted in hematological parameters (i.e., RBC or WBC counts), urinary and renal examinations, organ weights, cumulative death rates, mean body weights, or food intake. The highest dose at 50,000 ppm (reported as 2.5 g/kg bw-day by JECFA, 1974; EFSA, 2018) was identified as the NOAEL and served as the basis for EFSA's ADI (EFSA, 2018). Although Gaunt et al. (1972) did not observe any statistically significant changes in hemolytic effects at the end of their study, an increasing trend in reticulocytes was observed at the 52-week time-point, suggesting a possible compensatory response to RBC damage. A statistical trend analysis was not conducted.

In addition, a 90-day inhalation rat study at 321 and 707 ppm of PG was conducted and found decreases in WBC count among females at both concentrations (Suber et al., 1989). At 707 ppm, mean corpuscular hemoglobin concentrations also decreased, but only in females. No dose-related changes in red or white blood cells were observed in male rats at either exposure level.

Gaunt et al. (1972), the 2-year rat study, identified the highest NOAEL, which is below the lowest reliable LOAEL of 5000 mg/kg bw-day (from Weil et al. 1971). Gaunt et al. (1972) has served as the basis for the POD in other assessments (JECFA, 1974; EFSA, 2018). The POD is the NOAEL of 2500 mg/kg bw-day, which is higher than the NOAEL of 2000 mg/kg bw-day noted in the dog study (Weil et al., 1971). Additional studies that reported adverse, treatment-related effects evaluated potential toxicity following a single exposure application and were therefore not considered relevant as key studies for a chronic health-based toxicity value (Saini et al., 1996; Lau et al., 2012). Based on our review of the available scientific literature, we concluded that the NOAEL of 2500 mg/kg bw-day from Gaunt et al. (1972) represents the most appropriate POD.

3.2. human studies

Many human observational studies on PG (i.e., case reports of intensive care unit [ICU] patients, retrospective and prospective studies in a similar setting) are of limited value for determining a POD. Still, they may be useful in evaluating biological plausibility, developing a better understanding of PG toxicokinetics, and quantifying doses that elicit a toxicological response. The majority of human studies are case

Table 2

Relevant animal studies to potentially inform PODs.

Reference	Species	Route, Dose	Duration	Observed Effects	NOAEL (mg/kg bw-day)	LOAEL (mg/kg bw-day)	Klimisch Score ^a
Short-term and Sub-chronic Oral Studies							
Vaillie et al. (1971, as cited in US EPA, 2008)	Wistar rats (10 males/dose)	Drinking water, 0, 5 or 10% (equivalent to 0; 5200; or 10,000 mg/kg bw-day)	5 weeks	No gross or microscopic lesions in any organs examined. Decreased RBC count, increased liver weight, increased blood sugar, decreased blood urea.	NA	5200	K = 4 (Not assignable)
	Rabbits (7 males/dose)	Drinking water, 0 or 5% (equivalent to 4750 mg/kg bw-day)	8 weeks	No effects on body weight or blood sugar. Other endpoints not evaluated.	>4750	NA	
Saini et al. (1996)	Wistar rats. Study 1: 6 females/dose Study 2: 4 females/dose per duration group	Oral gavage, Study 1: 730 or 2940 mg/kg bw Study 2: 0, 730 or 2940 mg/kg bw	Single dose. Study 1: blood samples taken up to 7 days after exposure. Study 2: sacrificed at 30 min, 120 min, 2 days, or 7 days after exposure.	Decreased hemoglobin, packed cell volume, RBC counts; increased reticulocyte counts, plasma hemoglobin, osmolality, membrane surface changes at all PG exposures up to 2 days; increased spleen weights. By day 7, values returned to background levels.	NA	730	K = 2 (Reliable with restrictions)
Thackaberry et al. (2010)	CD-1 mice (10/sex/dose), Sprague-Dawley rats (10/sex/dose), Beagle dogs (4/sex/dose), Cynomolgus monkeys (4/sex/dose)	Oral gavage, 0 or 1000 mg/kg bw-day	92–93 days (mice and rats), 95–97 days (dogs and monkeys)	No adverse effects observed. In male mice, minimally elevated transaminase levels observed. No relevant histopathologic finding.	>1000	NA	K = 3 (Not reliable)
Chronic Oral Studies							
Gaunt et al. (1972)	Charles River CD rats (15/sex/dose)	Diet, 0; 6250; 12,500; 25,000; or 50,000 ppm (equivalent to 200–300; 400–500; 900–1000; or 2100–2500 mg/kg bw-day)	2 years	No effects on hematological parameters (i.e., RBC or WBC counts), urinary and renal examinations, organ weights, cumulative death rates, mean body weights, or food intake.	2500	NA	K = 2 (Reliable with restrictions)
Weil et al. (1971)	Beagle dogs (5/sex/dose)	Diet, 0; 2000; or 5000 mg/kg bw-day	2 years	Decreased hemoglobin, hematocrit, and erythrocyte counts; increased reticulocytes, anisocytes, and poikilocytes. No exposure-related pathological changes in bone marrow, spleen, liver or other tissues.	2000	5000	K = 2 (Reliable with restrictions)
Reproductive/Developmental Oral Studies							
Lau et al. (2012)	C57BL/6 mice (6–12/dose)	i.p., 0, 1, 2, 3, 5, and 10 mL/kg (equivalent to 0, 1000; 2000; 3000; 5000; 10,000 mg/kg bw) i.p., 10 mL/kg (equivalent to 10,000 mg/kg bw)	Single dose at PND 7 Single dose at PND 4, 7, 14, 24, or 30	Increased apoptosis in brain; caudate/putamen and cortical regions were most affected Increased apoptosis in brain	1000 NA	2000 10,000	K = 2 (Reliable with restrictions)
Environmental Health Research and Testing, Inc. (1985)	CD-1 mice (20/sex/dose)	Drinking water, 0, 1, 2.5, 5% (equivalent to 0; 1820; 4800; and 10,100 mg/kg bw-day)	18 weeks: 1 week prior to co-habitation, 14 weeks during co-habitation, 3 weeks after co-habitation.	In F0 generation, no effects on number of litters per pair, number of live pups per litter, sex ratio, pup weight, number of days to litter, dam weight at delivery, pup survival, or pup body weight gain. In F1 generation, no effects on reproductive parameters (organ weights, fertility index, litter size, number of live pups, sex ratio, pup weights, sperm parameters in males, estrus cycle in females).	>10,100	NA	K = 2 (Reliable with restrictions)
Kavlock et al. (1987)	CD-1 mice (30 pregnant females/dose; 40 controls)	Oral gavage, 0 or 10,000 mg/kg bw-day	GD 8–12	No effects on maternal (number pregnant, mortality, number of animals with resorptions) or fetal parameters (number of live pups, pup weight).	>10,000	NA	K = 2 (Reliable with restrictions)

Notes:

bw = Body Weight; US EPA = United States Environmental Protection Agency; GD = Gestational Day; i.p. = Intraperitoneal Injection; LOAEL = Lowest Observed Adverse Effect Level; NA = Not Available; NOAEL = No Observed Adverse Effect Level; PND=Postnatal Day; RBC=Red Blood Cell.

^a Data reliability assessed following the approach described by Klimisch et al. (1997).

Table 3Hematological findings from a 2-year dietary study in dogs, as reported in [Weil et al. \(1971\)](#).

Months	Males					Females					Reference Range ^a
	Control	2 g PG/kg-day	5 g PG/kg-day	2.54 g D/kg-day	6.35 g D/kg-day	Control	2 g PG/kg-day	5 g PG/kg-day	2.54 g D/kg-day	6.35 g D/kg-day	
Total Erythrocytes (10 ⁶ /mm ³)											
0	7.23	7.25	7.33	7.01	6.92	7.24	7.79	7.96	7.54	7.63	4.95-7.87
6	7.50	7.01	6.97	7.42	8.13	7.62	7.26	6.84	7.91	8.00	
12	7.28	7.19	6.68	7.54	7.44	7.62	7.61	7.07	7.89	7.31	
23	7.48	7.08	6.64	7.47	7.53	8.07	7.87	7.08	7.78	8.52	
23.5	7.47	6.53	6.40	7.39	7.46	7.21	6.98	6.25	7.67	7.66	
Hemoglobin (g/100 ml)											
0	15.1	14.6	15.0	14.3	14.6	15.2	16.3	17.2	15.9	15.6	11.9–18.9
6	18.0	16.7	16.7	17.7	19.1	17.7	17.3	16.3	18.5	18.3	
12	18.2	18.1	17.1	18.6	18.8	18.8	18.6	17.5	19.3	18.2	
23	18.8	17.4	16.8	18.6	18.7	20.1	19.2	17.8	18.9	20.1	
23.5	17.8	16.3	15.7	17.8	18.1	18.5	16.8	15.4	18.3	18.5	
Hematocrit (%)											
0	46.1	43.6	45.9	42.9	44.6	45.5	48.1	50.4	48.3	46.1	35–57
6	49.1	46.1	47.4	47.9	52.4	48.3	48.6	46.3	49.4	50.0	
12	48.1	48.1	46.7	49.4	50.0	49.9	49.9	47.9	50.9	46.9	
23	49.6	46.4	45.5	49.2	49.7	51.8	51.3	47.6	49.4	53.7	
23.5	47.6	42.0	43.2	47.7	48.1	47.3	46.3	42.7	48.0	48.8	
% Nucleated Erythrocytes											
23.5	3.6	2.8	4.4	2.8	2.6	1.8	3.0	8.0	2.0	1.6	NA ^b
% Reticulocytes											
24	1.80	1.55	2.08	1.32	1.44	1.32	2.04	3.36	1.50	1.44	0–1.0
Anisocytosis (Number of Dogs)											
23.5 normal	5	4	0	5	0	5	4	0	4	4	NA
23.5 slight	0	0	2	0	5	0	1	3	0	1	
23.5 moderate	0	0	3	0	0	0	0	2	0	0	
Poikilocytosis (Number of Dogs)											
23.5 normal	5	4	0	5	0	5	4	0	4	4	NA
23.5 slight	0	0	2	0	5	0	1	3	0	1	
23.5 moderate	0	0	3	0	0	0	0	2	0	0	

Notes:

Bold = statistically different from control value, $p < 0.05$.

D = Dextrose; NA = Not Available; PG = Propylene Glycol; WBC=White Blood Cells.

^a Source: [Merck & Co., Inc. \(2023\)](#).^b Standard reporting of nucleated erythrocytes or reticulocytes is as %RBCs.

reports of ICU patients; in these studies, PG is present as an excipient, typically for lorazepam, and administered *via* bolus or continuous intravenous (IV) infusions ([European Medicines Agency, 2017](#)). A survey of the available literature finds that almost all of these case studies involve IV doses of PG > 500 mg/kg bw-day, with many patients receiving doses >1000 mg/kg bw-day ([Table 4](#)). An early sign of PG intoxication includes increases in serum PG, which can result in serum hyperosmolality and an elevated osmol gap (see MoA section for further discussion of hyperosmolality and osmol gap). Both of these indicators measure the amount of excess plasma solute and are common biomarkers of overexposure to alcohols ([Greene and Krasowski, 2020](#)). Increases in PG serum concentration and osmol gap were observed consistently across case studies. In the case reports, clinical effects reported among adults include lactic acidosis, intravascular hemolysis, kidney effects, and hemoglobinuria ([NTP, 2004](#); [European Medicines Agency, 2017](#)). However, because of the high doses characteristic of case studies and because of severe and large variability in underlying disease status in the cases (e.g., kidney disease, alcohol abuse, respiratory failure), case reports are of limited value in establishing a dose-response relationship between PG exposures, early biomarkers, and any possible adverse effects ([European Medicines Agency, 2017](#)).

Several available retrospective and prospective studies provide further perspective on the dose-response relationship between PG exposure and downstream effects. Similar to the case reports, these studies also have significant limitations. PG doses and exposure times vary widely across human studies, as does the timepoint for measurement of PG serum levels and other parameters (e.g., osmol gap). The

latter inconsistency (in time at which measurements are taken) is especially problematic when considering the short half-life of PG and how rapidly PG serum C_{max} is reached ([Yu et al., 1985](#)). Additionally, pre-existing conditions common in many ICU patients can affect the toxicokinetic profile (e.g., kidney disease). Indeed, several studies have found that PG serum levels are correlated with baseline kidney function ([Horinek et al., 2009](#)) or APACHE II score, which uses several biological parameters to estimate mortality potential in critically ill individuals ([Nelsen et al., 2008](#)). In addition to these limitations, the available prospective and retrospective observational studies often involve a small number of study participants, lack controls, administer PG doses that are not sufficiently characterized, and, like the case studies, involve critically ill ICU patients with a host of serious underlying health conditions. Because of these limitations, results among studies are highly variable and it is difficult to establish a "safe level" PG exposure ([Wilson et al., 2005](#)). Nonetheless, several key observations can be made:

- A consistent relationship between PG exposure, PG serum increases, and increased osmol gap can be observed in many studies (e.g., [Arroliga et al., 2004](#); [Horinek et al., 2009](#); [Yaucher et al., 2003](#)). The strength of this relationship is dependent on the magnitude of the dose, rate of infusion, and how recent the exposure occurs. For example, [Arroliga et al. \(2004\)](#) found a correlation between mean cumulative high dose PG (~1600 mg/kg bw-day) over a 48-h period, but not lower dose cumulative PG (~1000 mg/kg bw-day) occurring over a longer time frame (mean: 5100 mg/kg/4.9 days). Also, [Horinek et al. \(2009\)](#) observed that a high continuous infusion rate

Table 4

Case reports of intravenous administration of propylene glycol and effects on serum PG, osmolality, and osmol gap.

Age (yrs)/ Sex	Body Weight (kg)	Infusion Rate of PG (g/hr) ^a	Average/ Maximum Infusion Rate	Duration of Therapy (days)	Total Cumulative Dose of PG (g)	Estimated Average Daily Dose (mg PG/kg bw-d)	Serum PG Conc. (mg/dL)	Serum Osmolality (mOsm/kg)	Osmol Gap	Reference
46/M	208	3.25	maximum	12	552	221 ^b	30	346	15	Hayman et al. (2003)
39/M	91.7	1.74	maximum	NA	301	457 ^c	5	NA	3	Yahwak et al. (2008)
59/M	70 ^d	1.5	average ^e	7	249	508 ^b	74.5	353	31	Yaucher et al. (2003)
41/F	95.3	2.31	maximum	NA	160	581 ^c	55	NA	7	Yahwak et al. (2008)
69/M	63.6	1.65	maximum	NA	165	623 ^c	0	NA	6	Yahwak et al. (2008)
39/M	70 ^d	1.8	average ^e	9	398.4	632 ^b	26.2	352	33	Yaucher et al. (2003)
32/F	150	4.15	maximum	NA	202	664 ^c	28	NA	12	Yahwak et al. (2008)
29/M	70 ^d	2.0	average ^e	75	3685.2	702 ^b	24	326	4	Yaucher et al. (2003)
72/M	76.5	2.25	maximum	NA	316	706 ^c	8	NA	0	Yahwak et al. (2008)
62/M	107.1	3.52	maximum	NA	345	789 ^c	43	NA	10	Yahwak et al. (2008)
48/M	70 ^d	2.4	average ^e	15	846.6	806 ^b	254.4	386	64.2	Yaucher et al. (2003)
67/F	70 ^d	2.4	average ^e	14	796.8	813 ^b	77.9	316	12.3	Yaucher et al. (2003)
60/M	70 ^d	2.49	maximum	>5	540	854 ^c	78	321	21	Arbour and Esparis (2000)
48/M	70 ^d	2.6	average ^e	7	444	906 ^b	144	405	NA	Wilson et al. (2005)
35/F	56.1	2.13	maximum	NA	200	913 ^c	30	NA	13	Yahwak et al. (2008)
48/M	84	3.20	maximum	NA	175	913 ^c	25	NA	11	Yahwak et al. (2008)
31/M	94.3	3.59	maximum	NA	87	913 ^c	2	NA	10	Yahwak et al. (2008)
77/M	70 ^d	2.7	average ^e	27	1768	935 ^b	18.6	316	12.2	Yaucher et al. (2003)
15/M	56	2.33	maximum ^f	42	NA	999 ^c	NA	331	25	Seay et al. (1997)
38/F	70 ^d	3.0	average ^e	7	498	1016 ^b	61.5	317	24	Yaucher et al. (2003)
34/F	86	5.41	maximum	3	284	1101 ^b	12	345	24	Cawley (2001)
81/M	68.1	3.30	maximum	NA	465	1162 ^c	70	NA	26	Yahwak et al. (2008)
52/M	70 ^d	3.5	average ^e	9	747	1186 ^b	345	356	49	Yaucher et al. (2003)
49/M	104	6.11	maximum	NA	423	1411 ^c	95	NA	12	Yahwak et al. (2008)
63/F	44.1	2.82	maximum	NA	585	1536 ^c	100	NA	12	Yahwak et al. (2008)
53/M	70 ^d	5.4	average ^e	7	899	1835 ^b	NA	408	NA	Wilson et al. (2005)
41/M	71	4.79	maximum	NA	223	1619 ^c	3	NA	5	Yahwak et al. (2008)
62/M	59 ^g	4.15	average	4	398.4	1688 ^h	382	NA	81.4	Tsao et al. (2008)
34/F	70 ^d	9.96	maximum	24	2998	1785 ^b	520	380	97	Parker et al. (2002)
55/F	55.1	4.19	maximum	NA	1378	1826 ^c	110	NA	34	Yahwak et al. (2008)
59/F	90	7.2	maximum ^e	1 (24 h)	173.7	1930 ^{b,i}	170	322	34	Miller et al. (2008)
67/F	50	4.15	average	5	498	1992 ^h	250	395	89.4	Yan et al. (2010)
34/M	70 ^d	6.4	average ^e	2	306	2186 ^b	175	336	NA	Wilson et al. (2005)
45/M	70 ^d	10.79	maximum	38	5983	2249 ^b	17	355	48	Arbour (1999)
56/F	75	17.9	average ^e	<1 (10 h)	179.28	2390 ^b	457	NA	32	Tuohy et al. (2003)

(continued on next page)

Table 4 (continued)

Age (yrs)/Sex	Body Weight (kg)	Infusion Rate of PG (g/hr) ^a	Average/Maximum Infusion Rate	Duration of Therapy (days)	Total Cumulative Dose of PG (g)	Estimated Average Daily Dose (mg PG/kg bw-d)	Serum PG Conc. (mg/dL)	Serum Osmolality (mOsm/kg)	Osmol Gap	Reference
36/M	70 ^d	7.05	maximum	8 (non-consecutive)	NA	2419 ^c	NA	342	27	Reynolds et al. (2000)
60/F	70 ^d	7.1	average ^e	3	511	2433 ^b	588	318	41	Wilson et al. (2005)
41/M	70 ^d	7.6	average ^e	25	4565	2609 ^b	NA	326	NA	Wilson et al. (2005)
61/M	70 ^d	8.1	average ^e	5	970	2771 ^b	108	386	NA	Wilson et al. (2005)
28M	60 ^j	9	maximum	>3	NA	2918 ^{c,j}	130.8	344	38	Al-Khafaji et al. (2002)
30/F	70 ^d	8.9	average ^e	9	1912	3035 ^b	NA	384	NA	Wilson et al. (2005)
NA	70 ^d	8.875	average	7	1699	3043 ^h	NA	NA	145	Zar et al. (2007)
35/M	70 ^d	10.2	average ^e	2	490	3500 ^b	NA	326	35	Wilson et al. (2005)
33/F	60	21.15	average ^e	1 (29 h)	507.6	8460 ^b	NA	422	102	Ganesh and Audu (2008)

Notes:

bw = Body Weight; F = Female; g/hr = grams per hour; hrs = Hours; kg = kilograms; M = Male; mg/dL = milligrams per deciliter; mg PG/kg bw-d = milligrams of propylene glycol per kilogram of bodyweight per day; mOsm/kg = milliosmole per kilogram; NA = Not Available; PG = Propylene Glycol.

^a When infusion rate of PG was not reported, the rate for lorazepam (LZ) was converted assuming 1 mg LZ = 415 mg PG.

^b Calculated based on total cumulative dose of PG (g), body weight (kg), and treatment duration (days).

^c Calculated as the average daily dose at the maximum infusion rate.

^d Default body weight assumption.

^e Estimated infusion rate based on total dose of PG (g) and treatment duration (days).

^f Based on reported rate of 2.25 mL PG/hr and assuming a density of 1.0375 g PG/mL (Arroliga et al., 2004).

^g Calculated based on dose (per kg) provided in original study.

^h Calculated based on the reported infusion rate (g/hr), body weight (kg), and treatment duration (days).

ⁱ PG exposure presented here associated with pentobarbital administration, although additional medications containing PG were given during the treatment period.

^j Calculated based on reported max value of 108 g/m²-day and K_m of 37 kg/m², which corresponds to 60 kg body weight (US FDA, 2005).

and cumulative 24-h dose were strongly associated with PG serum levels, but total cumulative dose and duration were not. The lack of relationship is consistent with observations noted in the case reports. As presented in Table 4, the patient with the longest exposure to PG (75 days) and one of the highest cumulative doses (3685.2 g) had one of the lower serum osmolarities and osmol gaps. In contrast, the case study with the highest osmol gap had the highest infusion rate and was exposed for only one day.

- Not all studies report a consistent relationship between PG exposure, PG serum increases, and increased osmol gap. For example, Nelsen et al. (2008) did not find a statistically significant relationship between serum PG concentrations and osmol gap concentrations. The lack of an association between PG serum concentration and osmol gap appeared to be driven by a lack of association at low PG serum concentrations (see Figure S3 in Nelsen et al., 2008). This is consistent with toxicokinetic studies demonstrating that PG exposure and PG serum concentration follow first order kinetics at low dose and zero order kinetics at higher doses (Speth et al., 1987; Yu et al., 1985).
- The observations of metabolic disturbances (i.e., lactic acidosis, decrease in serum bicarbonate), kidney effects (e.g., creatinine increases, acute kidney failure), and other health effects are inconsistently observed across the observational studies and often difficult to attribute to PG exposure, likely due to the multi-faceted medical histories and compromised health statuses of study participants. For example, in Yahwak et al. (2008), the relationship between low-dose PG exposure (0–78 mg/kg bw-day PG) and creatinine increases were unclear (and unrelated to osmol gap). In contrast high-dose exposure (mean dose 620 mg/kg bw-day) caused more consistent manifestations of metabolic dysfunction and increased creatinine. While the study authors could not attribute these effects with certainty to PG exposure, the abnormalities resolved with the cessation of PG

exposure. As another example, Yaucher et al. (2003) found that serum creatinine increased in patients receiving 510–1190 mg/kg bw-day of PG. Out of 21 subjects receiving PG, Wilson et al. (2005) determined that four subjects (with PG serum levels ranging from 58 to 127 mg/dL) developed some metabolic abnormalities but the disturbances were reversible and did not result in clinical deterioration. In contrast, Nelsen et al. (2008), Barnes et al. (2006), and Horinek et al. (2009) found no relationship between PG exposure and any form of PG toxicity.

- Most human studies do not report hemolytic effects, although the repeated exposure studies evaluated doses lower than 5000 mg/kg bw-day (see Table 4). Some exceptions include the onset of hemolysis in patients at a PG bolus dose of 54 mg/kg bw (in an etomidate solution) over 30 s likely due to high serum osmolality (Doenicke et al., 1997). Infants, and particularly preterm infants, are more susceptible to PG toxicity because of a reduced ability to metabolize PG and reduced kidney clearance (NTP, 2004; European Medicines Agency, 2017). Hemolysis was reported in a 9-year-old patient who received IV propylene glycol in etomidate at a dose of ~0.4 g/kg-hr for 12 h (~4800 mg/kg bw/day) (Van de Wiele et al., 1995 as cited in European Medicines Agency, 2017). The ability of hyperosmotic PG solutions to cause hemolysis has also been observed in experiments involving IV bolus injection (Doenicke et al., 1997), and *in vitro* experiments (Jay and Rowlands, 1975). Importantly, serum PG concentrations in humans do not typically reach the PG concentrations measured in the dog study by Weil et al. (1971) (100 and 200 mg/dL, with C_{max} of 560 mg/dL), which could explain the absence of effect in most human studies.
- Although the current human data are not adequate to identify a threshold serum PG level associated with toxicity (Wilson et al. 2005), some studies have suggested PG serum levels of 18 mg/dL and greater are associated with PG toxicity (Yahwak et al., 2008), while

others have used a serum level of 25 mg/dL as a potential threshold for PG toxicity (Barnes et al., 2006; Horinek et al., 2009; Nelsen et al., 2008). As such, it is informative to calculate the dose of PG associated with the commonly utilized threshold of 25 mg/dL, using the regression equations derived from human studies. These studies found statistically significant relationships between the dose of the PG-containing drug, lorazepam, and serum PG levels. PG doses of approximately 370–650 mg/kg bw-day would result in a serum PG level of 25 mg/dL based on equations provided by Horinek et al. (2009) and Arroliga et al. (2004). This range is consistent with doses (>500 mg/kg bw-day) observed in case studies. Using 18 mg/dL as a more conservative threshold, the associated PG dose would be $\geq$ 250 mg/kg bw-day, assuming linearity.

To summarize, increased osmol gap in response to PG exposure is a key parameter measured in most human studies such as Barnes et al. (2006), Yahwak et al. (2008), and Yaucher et al. (2003).

3.3. Toxicokinetics

In general, absorption of PG – a small, highly water-soluble compound – from oral exposures is rapid and is primarily confined to the intravascular space (not tissues). For example, Yu et al. (1985) found that following oral exposure to 0.3–0.6 g/kg bw in adults, the maximum plasma concentration (C_{max}) of PG was reached within 1 h. Although quantitative oral bioavailability estimates are not reported in the literature, the available toxicokinetic information supports that PG is likely rapidly absorbed. Human toxicokinetic studies of PG using oral, IV, and rectal exposures report apparent volume of distribution (V_d) values ranging from 0.52 to 0.93 L/kg (Speth et al., 1987; Yu et al., 1985; Kolloffel et al., 1996), further supporting that PG is confined primarily to the intravascular space. Once absorbed, PG is metabolized to lactaldehyde by alcohol dehydrogenase and then to lactate by aldehyde dehydrogenase (NTP, 2004). Alcohol dehydrogenase activity is the saturable, rate-limiting step for PG metabolism with first-order kinetics at doses below metabolic saturation of $\sim$ 200 mg/kg bw and zero-order kinetics at doses above saturation (NTP, 2004; Speth et al., 1987).

PG has a relatively short half-life and can be eliminated both in its unchanged form or as a glucuronic acid conjugate from the kidneys, though it is not clear from the literature the relative contribution of these different forms to total body PG clearance (NTP, 2004). Toxicokinetic studies in humans administered PG orally or through IV reported estimated half-life values ranging from 1.6 to 5.6 h (Kelner and Bailey, 1985; Speth et al., 1987; Yu et al., 1985). The metabolic process is saturable and dose-dependent, resulting in increasing half-life and decreasing clearance values with increasing doses (Yu et al., 1985; Speth et al., 1987). Dose-normalized areas under the curve (AUCs) increased with increasing doses, indicating PG clearance follows nonlinear (rather than linear) elimination kinetics (Speth et al., 1987).

3.3.1. Interspecies toxicokinetics

Overall, PG absorption, distribution, metabolism, and excretion are similar across species (NTP, 2004; European Medicines Agency, 2017; EFSA, 2018). EMA (2017) concluded, "The absorption, distribution, metabolism and excretion of PG have been studied in human and in the animal species used in the toxicity studies (rat, mouse, dog, and monkey). No major differences were noted between pharmacokinetics in these species supporting the clinical relevance of their use in the assessment of the toxicology profile of propylene glycol. Despite the overall similarities, there is some uncertainty regarding the dose at which metabolism is saturated (an important feature in PG-induced toxicity).

Rat and rabbit studies reported rapid PG absorption *via* the oral route, with C_{max} being reached at 1 h in rabbits given 2.9 g/kg bw, and at 10 min or 2 h in rats given 0.4 g/kg bw or 5.9 g/kg bw, respectively (Morshed et al., 1988, 1991). Toxicokinetic studies indicate that

saturation of alcohol dehydrogenase (ADH) by PG appears to occur at lower doses in humans compared to rats and rabbits (Speth et al., 1987; NTP, 2004). Although the half-life data provided by Speth et al. (1987) suggest that PG metabolic saturation in humans occurs at lower doses than in rats and rabbits, these estimates are not reliable given the following uncertainties: metabolic clearance at doses lower than 5.1 g/day ($\sim$ 70 mg/kg bw-day), the lowest dose tested, cannot be estimated; the low number of subjects and the extensive interindividual variation precludes discerning with confidence the dose at which metabolic saturation might occur; and the appropriateness of comparing IV doses (and different dose schedules) in humans to those in rats and rabbits. Importantly, potential species differences in renal elimination should also be considered. PG renal clearance in humans has not yet been quantified in experimental studies.

The toxicokinetics of PG in rats and dogs following inhalation exposures (for up to 28 days) provide further support for the saturation of systemic PG metabolism and elimination, with other species differences also noted (Werley et al., 2011). In both species, absorption *via* inhalation appeared to occur *via* first-order kinetics, with C_{max} values in the plasma and lung roughly proportional to the nominal dose levels. In contrast, the AUC values from both the lung and plasma increased at supralinear levels. Zero-order elimination in rats was evident at the highest dose only (2160 mg/kg bw-day) and not at the next lowest dose of 720 mg/kg bw-day. Similarly, nonlinear elimination kinetics was noted in dogs, although saturation was evident at the highest nominal dose of 300 mg/kg bw-day, which is substantially less than the level observed in rats (2160 mg/kg bw-day). The authors noted that systemic exposures (C_{max} , AUC $_{\infty}$) in dogs were orders of magnitude higher than in rats, presumably due to less efficient hepatic metabolism. Little to no PG accumulation was observed in either species up to the highest doses.

While species sensitivity in metabolism is often a focus in deriving CSAFs, differences in renal clearance may also contribute to potential interspecies differences in toxicokinetics. Some portion of PG is detoxified through glucuronidation, a phase II metabolic process that enhances renal clearance. It is important to note that cats, as a species, lack the ability to generate the glucuronic acid conjugate of PG, which presumably results in higher effective serum concentrations of PG. This is consistent with the toxicological data that demonstrates cats are the most sensitive species to PG-induced hematotoxicity (LaKind et al., 1999). In contrast, dogs can clear 12–45% of circulating PG unchanged *via* the kidneys (Lehman and Newman, 1937). Several species-specific values for percent PG renal clearance exist, but experimental data for humans were not available.

Overall, the available toxicokinetic data are inadequate to derive a CSAF to account for potential interspecies differences. Despite the metabolic saturation in all studied species (*i.e.*, humans, rats, rabbits, and others), a lack of reliable quantitative information precludes our ability to discern the dose at which metabolic saturation occurs and whether this would result in appreciable interspecies differences in systemic PG concentrations. There is also uncertainty regarding the proportion and form(s) of PG cleared through the kidneys (NTP, 2004). For these reasons, the default interspecies toxicokinetic uncertainty factor (AK_{UF}) of 4.0 was used.

3.3.2. Intraspecies toxicokinetics

Infant sensitivity to PG toxicity is well established as infants have reduced metabolism and renal clearance compared to adults due to their lower liver ADH activity and immature renal function (Pikkarainen and Raiha, 1967; Alcorn and McNamara, 2002). Consistent evidence shows infants metabolize and excrete PG at a slower rate than adults. As noted previously, metabolic saturation in humans may occur at 200 mg/kg bw-day or above. Compared to 2–4 h in adults (NTP, 2004), a longer PG half-life is described in two studies in infants exposed *via* dermal application (16.9 h, calculated assuming first order kinetics) or IV administration (19.3 h) (Glasgow et al., 1983; Fligner et al., 1985). Studies reporting reduced PG clearance in infants are constrained by

confounding factors, e.g., premature or other health problems but a longer half-life for infants can be reasonably anticipated based on reduced metabolism and renal clearance (European Medicines Agency, 2017; NTP, 2004).

In contrast, children appear to metabolize and excrete PG similarly to adults after approximately 5 years of age due to maturation (and subsequent increase) of ADH activity (Kolloff et al., 1996). In Kolloff et al. (1996), various toxicokinetic parameters were measured, including C_{max} and T_{max} , half-life, apparent volume of distribution, and clearance. C_{max} and T_{max} values were refined using a linear curve-fitting program; however, the small sample size (children 5–12 years, $n = 4$) was considered a limitation. Reported values overlapped with previously reported toxicokinetic parameters in adults (Speth et al., 1987), supporting the notion that ADH maturation occurs by 5 years of age and immature ADH contributes to infants' sensitivity to PG effects.

Due to the age-dependent differences in metabolic and renal clearance function, the default intraspecies toxicokinetic uncertainty factor (HK_{UF}) of 3.16 was retained.

3.4. Toxicodynamics

Hemolytic effects that result from PG exposure are dose- and route-dependent (i.e., bolus vs. diet). Saini et al. (1996) reported reversible hematologic effects in rats administered single PG doses of 730 or 2940 mg/kg bw via gavage. Changes in hematologic parameters were not observed in mice given similar dietary doses for 2 years. In humans, the onset of hemolysis has been observed starting at a PG dose of 54 mg/kg bw-day when given as an IV bolus (in an etomidate solution) over 30 s to adults (Doenicke et al., 1997). An increase in free hemoglobin was observed as early as 5 min post-exposure. In another study, PG doses of 701 mg/kg bw-day or 1185 mg/kg bw-day (assuming 70 kg body weight) were administered IV over either 75 days or a 9-day period, respectively (Yaucher et al., 2003). The lower dose was associated with a PG serum level of 25 mg/dL and an osmol gap of 4 mOsm/kg, and the higher dose was associated with a PG serum level of 345 mg/dL and an osmol gap of 49 mOsm/kg (Yaucher et al., 2003).

The relationship between PG and changes in hyperosmolar serum concentration is well-established in both animal and human studies (Weil et al., 1971; Yahwak et al., 2008; Zar et al., 2007). Serum osmolality is the concentration of ions and particles dissolved in body fluid. It is primarily determined by sodium, potassium, their corresponding anions (chloride and bicarbonate), and uncharged species such as glucose and urea (Dorwart and Chalmers, 1975; Gennari, 1984). It is closely regulated and maintained within a tight range at about 275–295 mOsm/kg in humans. Serum PG levels in drug formulations and osmolality are directly related, with osmolalities of commercially available drug preparations (using PG as an excipient) ranging from 365 mOsm/kg to 12,800 mOsm/kg, with most measured at above 1000 mOsm/kg (Denicke et al., 1992).

Osmol gap is the difference between measured osmolality and calculated osmolality, and is used clinically as a screening test for the abnormal presence of unmeasured osmotically-active substances (Pursell et al., 2004). An osmol gap greater than 10 mOsm/kg may indicate the presence of various alcohols (including PG) and their metabolites in serum (Pursell et al., 2004). An osmol gap may be detected in patients with a variety of conditions (e.g., diabetes, alcoholic ketoacidosis, renal failure) or those receiving certain medical treatments.

Osmoregulation is a critical physiological process that serves to maintain RBC homeostasis, hemodynamics, and blood flow characteristics to ensure adequate microcirculation and tissue perfusion (Varga et al., 2022; Baskurt and Meiselman, 2003). Small changes in osmolality (i.e., $\pm 10\%$) can directly affect the function, shape, and normal properties of RBCs, resulting in not only changes in the cellular surface-to-volume ratio but also in intracellular viscosity and density as a direct result of swelling or shrinkage (Varga et al., 2022). A hyperosmolar environment will cause RBC shrinkage due to water outflow,

resulting in spinocytes (characteristically shaped RBCs). PG exposure has also resulted in hemolysis at high concentrations *in vitro* (Randolph and Mallery, 1944; Cadwallader, 1963; Cadwallader et al., 1964; Reed and Yalkowsky, 1985), due to hyperosmolarity from increased ion concentrations outside of erythrocytes (Demey et al., 1988). Plasma hyperosmolarity has also been shown to enhance membrane cross-bonding and cellular adherence in RBCs, which may contribute to RBC damage and removal by altering RBC surface characteristics (Saini et al., 1996). A compensatory increase in reticulocytes would be expected after PG-induced hemolysis (Bessman, 1990).

3.4.1. Interspecies toxicodynamics

Dog, rat, and human RBCs respond to a hyperosmolar environment in a similar manner, which is supported by generally similar species-specific ranges of normal blood osmolality (Table 5). In humans, normal plasma osmolality is maintained at 275–295 mOsm/kg, while the plasma osmolality of other mammals is slightly higher. Based on elongation index maximum (EI_{max}) values (i.e., the blood osmolality at the highest elongation index, which is a measure of RBC deformability), Varga et al. (2022) grouped different mammalian species into three clusters: human and dog; mouse, rat, and rabbit; pig, cat, and sheep. Humans and dogs were reported to have the lowest osmolality at EI_{max} , followed by rats, mice, and rabbits. However, it should be noted that the mean osmolality at EI_{max} for humans and rats is more similar (a mean difference of 12.25 mOsm/kg) than the mean osmolality at EI_{max} for humans and dogs (a mean difference of 31.05 mOsm/kg) (see Table 6).

The effects of hypertonic shock and hypotonic stress on RBCs are generally similar between humans, rats, and dogs. Shpakova and Orlova (2020) examined the osmotic stability of RBCs across various species under conditions of increasing salt concentrations and noted similar concentrations (i.e., ~ 2.75 mol/L NaCl) between humans and dogs provided the conditions for hypertonic shock of RBCs. In contrast, rat RBCs withstood conditions with slightly higher NaCl concentrations at ~ 3.0 mol/L (Shpakova and Orlova, 2020). Conversely, using the osmotic fragility index (i.e., the NaCl concentration at which 50% hypotonic hemolysis is detected), Shpakova and Orlova (2020) demonstrated that equine and bovine RBCs were most sensitive to hypotonic stress compared to human, dog, and rat RBCs (Shpakova and Orlova, 2020). The response of RBCs to osmolality changes was similar in humans and dogs, influenced only by the osmolality of the outside media (Rich et al., 1968).

While cats are most sensitive to PG effects on RBCs, they are not a good model for humans because cats do not have the ability to metabolize PG to its main glucuronidated metabolite, which prevents renal elimination of PG, increases PG levels in systemic circulation, and increases observed hematotoxicity of PG (NTP, 2004). For example, PG oral exposure to 1.2, 1.6, 2.4, or 3.6 g/kg bw-day for 2, 5, or 17 weeks in cats resulted in increased numbers of RBCs with Heinz bodies, composed

Table 5
Interspecies differences in normal and EI_{max} osmolality.^a

Species	Normal Osmolality Range (mOsm/kg)	Mean ($\pm$ SD) Osmolality at EI_{max} (mOsm/kg) ^b
Human	275–295	303.05 $\pm$ 10.92
Dog	280–300	272 $\pm$ 6.72
Rat	300–310	315.3 $\pm$ 18.45
Mouse	310–330	327 $\pm$ 14.57
Rabbit	310–330	319.75 $\pm$ 10.5
Cat	290–340	370.14 $\pm$ 19.94
Pig	310–340	360.88 $\pm$ 19.28
Sheep	310–340	396.88 $\pm$ 15.23

Note:

EI_{max} = Elongation Index Maximum; SD = Standard Deviation.

^a Source: Varga et al. (2022).

^b The authors did not report whether differences in osmolality at EI_{max} were statistically significant among species.

Table 6
Suggested revisions to the toxicodynamic CSAFs in PG ADI derivation.

ADI	POD (mg/kg bw-day)	Animal-to-Human Differences		Human Variability		Total UF ($AD_{UF} \times AK_{UF} \times HD_{UF} \times HK_{UF}$)	ADI (mg/kg bw-day)
		Dynamics (AD_{UF})	Kinetics (AK_{UF})	Dynamics (HD_{UF})	Kinetics (HK_{UF})		
EFSA (2018)	2500	2.5	4.0	3.16	3.16	100	25
Proposed	2500	<u>1.0^a</u>	4.0	3.16	3.16	40	<u>62.5</u>

Notes:

ADI = Acceptable Daily Intake; AD_{UF} = Interspecies Toxicodynamic Uncertainty Factor; AK_{UF} = Interspecies Toxicokinetic Uncertainty Factor; bw = Body Weight; CSAF = Chemical-Specific Adjustment Factor; EFSA = European Food Safety Authority; HD_{UF} = Intraspecies Toxicodynamic Uncertainty Factor; HK_{UF} = Intraspecies Toxicokinetic Uncertainty Factor; PG = Propylene Glycol.

^a This would be considered a chemical-specific adjustment factor (CSAF) rather than an uncertainty factor.

of denatured proteins such as hemoglobin (NTP, 2004). When mammals other than cats are exposed to PG, a fraction of PG is eliminated unchanged by the kidney and some is metabolized by the liver into lactate. In most mammals (but not cats), the remaining PG is conjugated with glucuronic acid and excreted in the urine (NTP, 2004). Thus, cats are not predictive of human PG response (LaKind et al., 1999).

Relative to interspecies differences in toxicodynamics, increased serum PG increases the osmol gap, stress on RBCs (leading to their destruction), and compensatory RBC production (as evidenced by increased reticulocytes). As the effect of increased serum osmolality on RBCs does not differ significantly between humans and other species (particularly, rats and dogs used in the toxicity studies), the CSAF interspecies toxicodynamic differences should be reduced from AD_{UF} of 2.5 down to 1.0.

3.4.2. Intraspecies toxicodynamics

Although there is some evidence that neonatal RBCs may respond differently to a hyperosmotic environment compared to adult RBCs (Linderkamp et al., 1983), there is little research linking PG-induced hyperosmolality to intraspecies differences in PG sensitivity. Because of age-related differences (from immature metabolism and renal clearance before 5 years of age), infants and those with immature or compromised kidney and liver function are likely to be more sensitive to PG's hemolytic effects (NTP, 2004). Since uncertainty with respect to the intraspecies toxicodynamics differences remains, the default uncertainty factor of HD_{UF} 3.16 was preserved.

3.5. Hypothesized MoA and revised ADI

Together, the evidence suggests a plausible MoA in which high PG serum levels lead to increased serum osmolality and an osmol gap and subsequent stress on RBCs (and their destruction), which may trigger a compensatory increase in RBC production (i.e., increased reticulocytes). The hemolytic effects of PG on erythrocytes, including decreased hematocrit and increased reticulocytes, have been observed in the 2-year dog study and in short-term rat studies. A high reticulocyte percentage or count suggests that the bone marrow is attempting to compensate for RBC destruction, or to recover from anemia (Bessman, 1990). This MoA is further supported by the osmotic hemolysis of human RBCs following exposure to PG *in vitro* (Jay and Rowlands, 1975), and increased free hemoglobin and decreased serum haptoglobin (presumably due to PG-induced hemolysis) in 49 patients upon a one-time injection of 0.3 mg/kg bw etomidate dissolved in PG (Doenicke et al., 1997).

In accordance with IPCS (2005) guidance,
POD:

- Gaunt et al. (1972) is the most reliable study in rats (dietary administration over 2 years). It has served as a basis for the POD in other key assessments, including EFSA (2018). The POD is the NOAEL of 2500 mg/kg bw-day in rats.

Toxicokinetics Differences

- *Interspecies*: There are species differences in PG metabolism and excretion (e.g., variable ADH activity and levels). Evidence supports metabolic saturation in humans, rats, rabbits, and other species. Due to the uncertainty associated with the wide variation in saturating metabolic levels and with the percentage of PG renal clearance in humans, the uncertainty factor for toxicokinetic interspecies differences was retained at AK_{UF} of 4.0.
- *Intraspecies*: Because of age-dependent differences in metabolic and renal clearance function (in particular, low levels of ADH activity in infants relative to adults), which affect the rate of PG elimination, the uncertainty factor for toxicokinetic intraspecies differences (HK_{UF}) was retained at 3.16.

Toxicodynamic Differences

- *Interspecies*: Increased concentrations of serum PG lead to increased osmol gap and stress on RBCs, RBC destruction, and compensatory increases in RBC production (as evidenced by increased reticulocytes). While EFSA (2018) used an AD_{UF} of 2.5, we suggest that the toxicodynamic interspecies CSAF be reduced to 1.0 because the effects of increased serum osmolality on RBCs are generally similar across species.
- *Intraspecies*: Although RBCs in neonates (compared to adults) may respond differently to a hyperosmotic environment (Linderkamp et al., 1983), little research links PG-induced hyperosmolality to intraspecies differences in toxicodynamics. The uncertainty factor for toxicodynamic intraspecies differences (HD_{UF}) was retained at 3.16.

Integrating this MoA, an increase of the ADI derived for PG from 25 mg/kg bw-day to 62.5 mg/kg bw-day is scientifically supported, with the POD based on the 2-year rat study NOAEL of 2500 mg/kg bw-day (Gaunt et al., 1972) and retaining all UFs except for the toxicodynamics interspecies differences, AD_{UF} , for which the CSAF was reduced from 2.5 to 1.0) (Table 6).

4. Discussion

PG has been used as an additive in foods, cosmetics, and pharmaceuticals for decades, in part due to its low toxicity. The most recognized concern with respect to PG toxicity is acute poisoning at extremely high doses. Long-term, repeat-dose studies (i.e., 2-year studies) demonstrate a very low hazard profile, showing no effects up to 2500 mg/kg bw-day in rats and minor effects up to 5000 mg/kg bw-day in dogs. Interpretation of human studies in ICU patients is challenging because of confounding factors, but the most widely observed effects were increases in osmol gap, lactic acidosis, and renal effects (Wilson et al., 2005; European Medicines Agency, 2017). Numerous studies have shown a rapid decrease in PG serum levels and osmol gap, as well as the return of any

other metabolic-related disturbances back to normal levels after PG withdrawal or hemodialysis (Parker et al., 2002; Barnes et al., 2006; Yaucher et al., 2003). The Agency for Toxic Substances and Disease Registry (ATSDR) reported no human deaths from PG exposures (Agency for Toxic Substances and Disease Registry, 1997).

Joint FAO/WHO Expert Committee on Food Additives, (2002) estimated a daily intake for PG of 2400 mg/day (equivalent to 34 mg/kg bw-day for a 70-kg adult); US EPA (2008) reported the per capita value as 14 mg/kg bw-day (based on US FDA estimates). These intakes are well below both the dose range of 400–1000 mg/kg bw-day that has been acknowledged as "safe" by human clinical publications (e.g., Zar et al., 2007; Wilson et al., 2005; Yahwak et al., 2008; Horinek et al., 2009) and the 500 mg/kg bw-day safe limit for adults according to EMA (2017). They are also below the dose associated with metabolic saturation, which has been estimated to be approximately 200 mg/kg bw-day (based on a change from first-order to zero-order kinetics at 12 g/day, NTP, 2004). Moreover, the daily intake estimate of approximately 34 mg/kg bw-day is a crude estimate based on annual production and assumes only 10% of the population are consumers (JECFA, 2002). More realistic estimates of average exposure, ranging from 0.67 to 11.6 mg/kg bw-day are available for the EU, Japan, and Korea (see Table 7). Using these values, the margin of exposure between general population PG intakes and dose associated with adverse effects would be even higher.

While limited, data on background PG serum levels provide further information on the large margin of safety between PG exposure in the general population and PG serum concentrations (~18–25 mg/dL) associated with adverse effects. Barnes et al. (2006) reported PG serum levels in the control group as "undetectable"; the control group had received a cumulative PG dose of 11 g but had not received any PG in the last 36 h. Interestingly, Burkhardt et al. (2021) documented background PG serum levels in the general German population (n = 10 in control group) below 0.3 µg/mL (or 0.03 mg/dL, estimated from Figure S3), which is over 600–800 times lower than a PG serum level of 18–25 mg/dL (Yahwak et al., 2008; Horinek et al., 2009).

While several recent authoritative assessments have investigated PG safety, none have sought to establish a plausible MoA for PG toxicity (EFSA, 2018; European Medicines Agency, 2017; NTP, 2004). As summarized in the Human Studies section, challenges with the human studies are that many ICU patients have a host of severe pre-existing conditions and co-exposures, or the human studies generally lack appropriate controls and reliable dose-response information. Most animal studies did not report long-term effects, and when effects were observed, they were minor and minimally adverse. More specifically, observed hematological effects in Weil et al. (1971) included decreased

total erythrocytes, decreased hemoglobin concentration, and decreased hematocrit, all within normal biological ranges. While the percent of reticulocytes was outside normal ranges, an increase in reticulocytes is an adaptive response to RBC damage (Merck & Co., Inc., 2023). Although the EFSA and JECFA suggest these effects may correspond to a LOAEL (Joint FAO/WHO Expert Committee on Food Additives, 1974; ECHA, 2018), EMA instead concluded that "... no real adverse propylene glycol-related effects appear to occur up to the highest dose in this study" (European Medicines Agency, 2017).

Collectively, the evidence supports a plausible MoA for the hematological effects of PG that entails an increase in PG serum levels, followed by an increase in the osmol gap, which at high enough levels can cause damage to RBCs. The increase in PG serum levels is a "key event" and has a threshold related to the metabolic saturation of ADH. At doses above this threshold, elimination changes from first-order to zero-order kinetics and PG starts to accumulate in the serum. To the contrary, PG does not accumulate if zero-order kinetics is not reached (Greene and Krasowski, 2020; Speth et al., 1987; Yu et al., 1985). Both EMA (2017) and NTP (2004) suggest that saturation occurs at approximately 12 g/day (~200 mg/kg bw-day); as such, doses under 200 mg/kg bw-day would not be expected to result in any PG accumulation or any other variations in metabolic parameters and therefore would not be expected to result in adverse effects in humans.

Doses of PG (≥200 mg/kg bw-day) over short periods are expected to overwhelm the metabolic capacity of ADH, rapidly elevating PG serum levels. As noted above, PG serum concentration and increased osmol gap were more strongly associated when exposures involved bolus doses over short periods of time, as opposed to lower doses over a longer exposure duration.

Some studies have suggested that the increase in serum osmolality may also be related to increased lactate concentrations, with reports of lactic acidosis in response to increases in PG exposure in several human studies (e.g., Yaucher et al., 2003; Arroliga et al., 2004). However, the results are inconsistent and are not definitively attributed to PG exposure and may relate to underlying health conditions (Barnes et al., 2006; Morris et al., 1942; Horinek et al., 2009). Regardless, increased serum lactate levels can also increase serum osmolality and impact RBCs, an effect not frequently observed in animal studies. Other studies have found that the metabolic pathways that convert PG to lactate in rats become saturated at approximately 2900 mg/kg bw-day, resulting in hyperlactatemia at 5900 mg/kg bw-day (Morshed et al., 1989). The dose associated with hyperlactatemia is higher than the highest dose in the 2-year dog study and could explain why an increase in serum lactate was not observed (Weil et al., 1971). Lactate increases were observed in at least one study in dogs that received PG in a dose of 4000 mg/kg bw-day activated charcoal (% PG not specified) (Burkitt et al., 2005).

In line with the EFSA (2018) opinion, we conservatively selected Gaunt et al. (1972) as the critical POD study, reporting a NOAEL of 2500 mg/kg bw-day (lower than the 5000 mg/kg bw-day level in dogs in which the small RBC effects noted were not clinically significant – all values within normal range – except for reticulocyte percent). Conversely, EMA utilized the NOAEL of 5000 mg/kg bw-day from the 2-year dog study when deriving its PDE for adults, noting, "... [n]o real adverse propylene glycol-related effects appear to occur up to the highest dose in this study" (European Medicines Agency, 2017). Had we opted to use the NOAEL of 5000 mg/kg bw-day as the POD in deriving a proposed ADI, applying IPSC guidance would have resulted in a 5-fold increase to the existing ADI (i.e., up to 125 mg/kg bw-day). This value, while not relied upon for this assessment, remains well below the human metabolic threshold of ~200 mg/kg bw-day and below doses associated with toxicity observed in the human clinical data. Our proposed revision to the ADI (62.5 mg/kg bw-day) relies on a lack of hematological changes in the rat, along with expected equivalency in effects across species that support the reduction of the CSAF for toxicodynamic interspecies differences from 2.5 to 1.0.

The reduction of the toxicodynamic portion of the interspecies

Table 7
Mean intake daily PG intake estimates.

Country	Mean (mg/kg bw-day)	Contextual Notes	Reference
EU	1.0–11.6	- Results for "brand-loyal" consumers, ≥12 months. - Estimates based on analytical measurements in food and specific food intakes from various EU member states.	EFSA (2018)
Korea	0.67	- Results for "consumers", all ages. - Estimates based on analytical measurements in food and specific food intakes from KNHANES data.	Lim et al. (2016)
Japan	0.71	- Study details not available.	Louekari et al., 1990 as cited in NTP, 2004

Notes:

bw = Body Weight; EFSA = European Food Safety Authority; EU = European Union; KNHANES = Korea National Health and Nutrition Examination Survey; PG = Propylene Glycol.

adjustment factor to 1.0 is consistent with agency assessments in which an UF of 1.0 was applied when a plausible MoA exists and the toxicological data indicate similar potency across test species (EFSA, 2019; US EPA, 2010). In US EPA's Integrated Risk Information System (IRIS) assessment of ethylene glycol monobutyl ether (EGBE), a chemical with many chemical similarities to PG, US EPA reduced the toxicodynamic uncertainty factor to 1 (US EPA, 2010). This adjustment was based on a study demonstrating that human RBCs are 40- to 150-fold less sensitive than rat RBCs to EGBE-induced hemolysis, which is a key step in the MoA. In another pertinent example, in the re-evaluation of phosphoric acid-phosphates, di-, tri-, and polyphosphates (EFSA, 2019), the toxicodynamic part of the interspecies UF was reduced to 1.0 because the key effect – calcification of the kidney and tubular nephropathy – is caused by the innate physio-chemical and physiological properties of phosphates. This is not unlike PG because PG's effect on RBCs is thought to be related to its contribution to total serum osmolality and not through interaction with a specific molecular target.

As discussed above, neonates are considered particularly sensitive to PG exposure, which is thought to be due to their immature renal function and metabolism (Alcorn and McNamara, 2002; Pikkarainen and Raiha, 1967 216–1999). Compared to 2–4 h in adults (NTP, 2004), a longer PG half-life is described in two studies of hyperosmolality in infants (Glasgow et al., 1983; Fligner et al., 1985). It should be noted, however, that studies reporting reduced PG clearance in infants are constrained by confounding factors (e.g., pre-existing conditions) and involved higher doses of PG than the adult studies, making direct quantitative comparisons between adults and infants uncertain (European Medicines Agency, 2017; NTP, 2004). After 5 years of age, children appear to metabolize and excrete PG similarly to adults because of maturing ADH activity (Kolloff et al., 1996). Importantly, Allegaert et al. (2010) observed no short-term toxic effects from a median PG exposure of 34 mg/kg bw-day in pre-term and low-birth-weight infants. Although the ADI does not apply to infants under 12 weeks of age, this suggests that PG exposures above the current ADI of 25 mg/kg bw-day could be tolerated even in a vulnerable subpopulation.

Based on the potential for children below the age of 5 years to have limited metabolic capacity, EMA established a PDE of 1 mg/kg bw-day for children under 5 years, and a higher safe exposure limit of 50 mg/kg bw-day for adults; this is informed by a mechanistic study in which a single intraperitoneal (IP) dose was administered and apoptosis in the central nervous system was noted (Lau et al., 2012). However, a single bolus IP dose would result in a different toxicokinetic profile from dietary administration studies at levels that would not exceed the PG metabolic capacity, as previously discussed. In addition, NTP (2004) concluded that PG is not a developmental toxicant based on its studies conducted in mice, rats, hamsters, and rabbits.

5. Conclusion

Animal studies have repeatedly demonstrated that PG has low toxicity, causing only non-adverse effects on RBCs at doses ≥ 5000 mg/kg bw-day. We have hypothesized that these effects are due to increases in serum osmolality resulting from PG serum accumulation upon metabolic saturation. Insufficient quantitative data precluded any refinements to the UF for toxicokinetics interspecies differences. Also, because individuals with underlying kidney disease, neonates, and infants may be more sensitive to PG toxicity (due to a decreased ability to metabolize and/or excrete PG), the current UF of 3.16 (to account for toxicodynamic intraspecies differences) remains appropriate. The toxicodynamics of the hyperosmotic environment on RBCs, however, are similar across species and support an interspecies toxicodynamics CSAF refinement from 2.5 to 1.0. This adjustment would support increasing the ADI from 25 to 62.5 mg/kg bw-day, providing an adequate margin of safety from higher doses that would be expected to overwhelm PG metabolizing capacity and lead to increased osmol gaps and hematological changes.

Funding source

ASL, SRB and CMM are employed by Gradient, a private environmental consulting firm. MMJ is Chief Science and Regulatory Officer at the American Beverage Association. The work reported in this paper was conducted during the normal course of employment. This work was funded by the American Beverage Association and the International Organization of the Flavor Industry. The authors have the sole responsibility for the writing, content, and conclusions of this article.

CRedit authorship contribution statement

Ari S. Lewis: Conceptualization, Writing – original draft, Writing – review & editing. **Steven R. Boomhower:** Conceptualization, Writing – original draft, Writing – review & editing. **Charlotte M. Marsh:** Conceptualization, Writing – original draft, Writing – review & editing. **Maia M. Jack:** Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ari Lewis, Steven Boomhower and Charlotte Marsh are employed by Gradient, a private environmental consulting firm. Maia Jack is Chief Science and Regulatory Officer at the American Beverage Association. The work reported in this paper was conducted during the normal course of employment. This work was funded by the American Beverage Association and the International Organization of the Flavor Industry. The authors have the sole responsibility for the writing, content, and conclusions of this article.

Data availability

No data was used for the research described in the article.

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